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Project Title:	Orientational ordering, pair interactions, and controlled self-assembly of magnetic nanowires in nematic liquid crystal solvents		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Liquid crystals		
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Task Description:

Colloids suspended in a complex fluid can be subject to a variety of interactions including electromagnetic, entropic, and gravitational forces. These interactions form the basis for many important topics in science and technology; two examples are the emergent field of microrheology and the self-assembly of colloidal particles. For colloids in a liquid crystal, the fluid's elastic energy provides a unique mechanism for interactions. This ground-based research program is an experimental study of orientational ordering and controlled self-assembly of magnetic nanowires suspended in liquid crystals. A colloidal particle in a liquid crystal matrix can introduce defects and other distortions in the nematic director due to the boundary conditions created by the anchoring of the director at the particle's surface. These distortions can, in turn, orient aspherical colloids with respect to the director as well as mediate anisotropic interactions between particles. Experiments are planned to characterize the orientational energy of wire-shaped colloids and their interactions within nematic liquid crystals for a variety of anchoring conditions. The use of magnetic particles allows controlled studies that include the application of torques and forces to the wires through external fields and field gradients that compete with the elastic forces of the nematic. Since the magnetic nanowires are potential building blocks for nanostructured devices, a particular focus of the work is toward creating conditions for self-assembly of the wires into ordered configurations that would facilitate the study of their electronic and magnetic properties or enhance their applications potential. The specific hypotheses driving the research strategy are: (i) close investigation of the nematic distortions around suspended wire-shaped colloids and the resulting interactions among the colloids can provide insightful perspectives on inclusions in liquid crystals, including the generation of defects; (ii) through judicious choice of surface treatments, the interactions mediated by nematic distortions can be tuned so as to form complex wire assemblies with useful properties; (iii) distortions in the nematic director field can be tailored to manipulate suspended wires, leading to complex assemblies with useful properties; and (iv) efforts to counter gravity can both affect nanowire behavior and lead to instrumentation development that impacts the study and application of magnetic colloids in fluids in general. To test these hypotheses, the PI's are building on their combined expertise in liquid crystal physics, electrodeposition for designing and fabricating magnetic nanowires, chemical techniques for functionalizing wire surfaces, and video microscopy for characterizing the behavior of suspended wires and the surrounding nematic. Because of the inherent density mismatch between the metal wires and the liquid crystal solvent, gravity strongly impacts the behavior of suspensions, and experiments are designed to illuminate its effects. This research has several specific objectives:
Objective 1: to determine the orientational forces imposed on isolated nanowires suspended in a nematic liquid crystal.

Objective 2: to characterize the dynamical behavior of isolated nanowires suspended in a nematic liquid crystal in response to controlled forces and torques.

Objective 3: to determine the interactions between pairs of nanowires and between nanowire and other objects mediated by nematic distortions.

Objective 4: to develop methods for the manipulation and self-assembly of nanowires that utilize the unique interactions experienced by nanowires in liquid crystals.

Objective 5: to modify surface properties of metal nanowires in order to control their nematic anchoring properties.

Objective 6: to investigate the effects of gravitational forces on nanowire interactions and self-assembly.

Objective 7: to develop instrumental tools that can mimic gravitational effects, and in particular can create effective low-gravity environments, for microscopy studies of magnetic nanowires in fluid suspension.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research conducted under this grant successfully addressed each of the objectives listed in the Task Abstract/Description. In addition, the grant enabled additional experiments on a closely related topic, the nonlinear microrheology of wormlike micelle solutions that display shear-induced nematic ordering. The research accomplishments fall roughly into seven major topics of experimental results and instrument development:
A) Temperature dependence of the orientational elastic energy of nanowires with planar anchoring in a nematic solvent.

Previously, we had determined that the elastic torque on a colloidal wire with longitudinal anchoring grows linearly with increasing angle between nematic director and wire's long axis, and hence the energy grows quadratically. These measurements were performed by balancing the elastic torque on a nickel wire suspended in the nematic with the magnetic torque from an external field. The observed behavior was observed to agree qualitatively with theoretical predictions of Brochard and de Gennes based on an analogy between the elastic energy and electrostatics. By tracking the torque as a function of temperature, we further showed that the agreement between the measured elastic energy and the theoretical prediction is remarkably quantitative.

B) Angular relaxation of nanowires with planar anchoring in a nematic solvent: a probe of anisotropic viscosity.

Due to the elastic torque, a wire initially at some angle with respect to the director will relax back to a parallel orientation. We characterized the dynamics of this process and showed how the wire's orientation as a function of time was a consequence of the nematic fluid's anisotropic viscosity. Further, by analyzing the wire motion when the orientation was near 90 degrees to the director and near parallel, we extracted effective viscosities that matched the nematic's Miesowicz coefficients, which correspond to the viscosities for simple shear flow along different high-symmetry directions. This result demonstrated the potential of the nanowires as microrheological probes, which we exploited further in the study of wormlike micelle solutions.

C) Orientation-dependent repulsive interaction with a wall.

We characterized the elastic interaction between a wire and a flat surface with planar anchoring. We observed an orientation-dependent repulsive force between wire and surface that we measured quantitatively by balancing the force with gravity. The results for the elastic energy agreed closely with calculations based on an extension of the electrostatic analogy of Brochard and de Gennes to this experimental geometry.

D) Self-assembly of magnetic manowires using patterned nematic anchoring on substrates.

Using lithographic techniques to pattern the nematic anchoring conditions on substrates, we generated periodic

Task Progress:	multi-domain nematic environments in which to investigate the manipulation and assembly of colloidal nanowires using the elastic forces of nematic solvents. This work focused on assembling wires in periodic twist/no-twist stripe-domain environments. The results of these studies showed that wires could indeed be assembled into patterns using the elastic forces from the nematic solvents and further that the forces experienced by the wires matched the expected values quantitatively. E) Wire-shaped inclusions with homeotropic anchoring. By employing a functionalization strategy that used mixed chain-length alkanes with carboxylate end groups to bond to the wire surfaces, we successfully modified the anchoring conditions at the surfaces of Ni nanowires to promote homeotropic (normal) anchoring of the neamatic director. We observed that these wires had a preferred alignment perpendicular to the director, consistent with theoretical expectations. We further observed the repulsive interactions between such wires and flat surfaces with planar anchoring. F) Nanowire microrheology of wormlike micelle solutions with shear-induced nematic order. As a way of exploring the potential of the nanowires for microrheology, we investigated the application of the nanowires to the nonlinear microrheology of wormlike micelle solutions, model non-Newtonian fluids that display a shear-induced transition to nematic order. These studies showed how the nanowires' characterization of linear viscoelastic behavior matches that of macroscopic measurements. Further, they showed how the correspondence between the microrheology and macroscopic shear rheology breaks down outside the linear regime, highlighting unique perspectives afforded by nonlinear microrheology as a result. In particular, the forces experienced by the wire revealed new aspects of the anisotropic nature of the fluid's nonlinear state and revealed directly the anisotropy of the drag in the nonlinear state. G) Design and fabrication of an 8-coil magnetic "tweezer" stage for optical microscopy. An important issue for magnetic colloids in complex fluids, such as the nanowires in nematics that were the focus research under this grant, is the density mismatch between metal particle and fluid and the resulting tendency for sedimentation. We designed and constructed an electromagnetic coil array for applying in situ magnetic fields in optical microscopy experiments that offset the effects of gravity on suspended magnetic particles. The design is optimized for the application of magnetic fields that are oriented in arbitrary directions within the microscope focal plane but that have gradients in their magnitude along the vertical. The gradients provide levitating forces to the suspended magnetic nanowires under observation, enabling them to be lifted controllably and held at arbitrary heights within a fluid while they are subjected to in-plane time-dependent fields. The capabilities afforded by this array were exploited extensively in the research under this grant, and we further anticipate that this design approach will have potentially wide application in the study of magnetic colloids in general.

Bibliography Type:	Description: (Last Updated: 04/30/2007)
Articles in Peer-reviewed Journals	Lapointe C, Cappallo N, Reich DH, Leheny RL. "Static and dynamic properties of magnetic nanowires in nematic fluids (invited)." Journal of Applied Physics. 2005 May 15;97(10):10Q304. http://dx.doi.org/10.1063/1.1852171 , May-2005
Articles in Peer-reviewed Journals	N. Cappallo, C. Lapointe, D. H. Reich, and R. L. Leheny "Nonlinear Microrheology of Wormlike Micelle Solutions Using Ferromagnetic Nanowire Probes" Physical Review Letters (submittted) , Apr-2007
Presentation	Leheny, R. L. "Elastic Torque and the Manipulation of Magnetic Nanowires in Anisotropic Complex Fluids" 49th Conference on Magnetism and Magnetic Materials, Jacksonville, FL Nov-2004
Presentation	Leheny, R. L. "Anisotropic Particles within Anisotropic Fluids" NIST Center for Neutron Research, Gaithersburg, MD Jan-2005
Presentation	Cappallo, N; Lapointe, C.; Leheny, R. L.; Reich, D. H. "Formation of localized shear induced states in wormlike micelle solutions using magnetic nanowire probes" 2005 APS March Meeting, Los Angeles, CA Mar-2005